

# PSNE of the Public Goods Game

$n$  players choose Contribute (C) or Not (N). Good provided iff at least  $k$  contribute.  
Preferences: free-ride  $\succ$  contribute+good succeeds  $\succ$  not contribute+ good fails  $\succ$  wasted capital.

Let  $s$  players choose Contribute, then we have the following cases

## Case 1: $s > k$ contributors

Any contributor can switch to N, still good provided ( $s - 1 \geq k$ ), and they go from outcome 2 to outcome 1. Strictly better. **Not a PSNE.**

## Case 2: $s = k$ contributors

- Contributor deviates to N:  $s$  drops to  $k - 1$ , good fails, outcome  $2 \rightarrow 3$ . No incentive.
- Non-contributor deviates to C: outcome  $1 \rightarrow 2$ . No incentive.

**PSNE.** There are  $\binom{n}{k}$  such equilibria.

## Case 3: $1 \leq s < k$ contributors

Good not provided. Any contributor gets outcome 4; switching to N gives outcome 3. Strictly better. **Not a PSNE.**

## Case 4: $s = 0$ (all N)

A lone deviator contributes,  $s = 1 < k$ , good still fails, outcome  $4 \rightarrow 3$ . No incentive. **Not a PSNE.**

## Result

All PSNEs: exactly  $k$  contributors ( $\binom{n}{k}$  equilibria) or nobody contributes (1 equilibrium). Total:  $\binom{n}{k} + 1$ .